Moving counting jumps disprove a periodic-oscillation conjecture

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Abstract

Give each nonempty binary word of length n the size $n + \sqrt{2}\,2^{-n}$. Its generating functions satisfy the analytic hypotheses of Bevan and Condé's Conjecture 6, including a full lattice of simple dominant poles. Nevertheless its cumulative count has no asymptotic of the asserted periodic form. This is a counterexample to MathDB 366299 and to the conjecture in the published article.

Task 1: a counterexample to the original conjecture

Conjecture 6 of [1], represented by MathDB 366299, predicts

$$|\mathcal{C}_{\le x}| \sim c(x)\rho^{-x}x^{\alpha-1} \qquad (x \longrightarrow \infty)$$

with bounded ω-periodic c, under two conditions: the irrational generating function has the decomposition

$$f(z) = g(z) + \frac{h(z)}{(1 - z/\rho)^{\alpha}}, \quad 0 < \rho < 1, \quad \alpha \notin -\mathbb{N},$$

where g, h are analytic on the closed cut disk $\overline{D}(0,\rho)\setminus\mathbb{R}_{\le 0}$; and the only singularities of the Dirichlet generating function on its convergence line are $\log(1/\rho) + 2\pi i\mathbb{Z}/\omega$. The counting variable x is real, as in the source's definition.

Theorem 1. *Let $\mathcal{C}$ consist of the nonempty binary words, with each word of length $n \ge 1$ assigned size*

$$\lambda_n = n + a2^{-n}, \qquad a = \sqrt{2}.$$

This is an irrational combinatorial class satisfying all these hypotheses with $\rho = 1/2$, $\alpha = 1$, $h = 1$, and $\omega = 1$. There is no bounded 1-periodic function c such that $|\mathcal{C}_{\le x}| \sim c(x)2^x$.

Proof. The class. There are 2^n words of size λ_n. Since $0 < a2^{-n} < 1$, we have $n < \lambda_n < n + 1$, so the sizes increase strictly, have no finite accumulation point, and have finite multiplicities. This is precisely the required notion of a combinatorial class. It is irrational even in the source's stronger sense of not having all sizes in any $\eta\mathbb{N}$, $\eta > 0$. Indeed, if $\lambda_1/\lambda_2 = r \in \mathbb{Q}$, then $1 + a/2 = r(2 + a/4)$; comparing the rational and irrational parts would give both $r = 1/2$ and $r = 2$.

The irrational generating function. Use the principal logarithm on $\mathbb{C}\setminus\mathbb{R}_{\le 0}$. Then

$$f(z) = \sum_{n\ge 1} 2^n z^{n+a2^{-n}}.$$

For fixed $z \ne 0$, the absolute value of its nth summand is $(2|z|)^n|z|^{a2^{-n}}$, with the last factor tending to 1. Thus the series converges for $|z| < 1/2$, and its terms fail to tend to zero for $|z| \ge 1/2$. Its radius is exactly $1/2$.

On $|z| < 1/2$ off the cut, subtracting a geometric series gives

$$\begin{aligned} f(z) &= \frac{1}{1-2z} + g(z), \\ g(z) &= -1 + \sum_{n\ge 1} 2^n z^n \big(\exp(a2^{-n}\operatorname{Log} z) - 1\big). \end{aligned}$$

We show that g is analytic on the larger open set $U = \{z : |z| < 1\} \setminus \mathbb{R}_{\le 0}$. On any compact subset of U, take $|z| \le r < 1$ and $|\operatorname{Log} z| \le L$. The elementary estimate $|e^w - 1| \le |w|e^{|w|}$ bounds the nth summand by

$$2^n r^n a 2^{-n} L e^{a2^{-n}L} \le aLe^{aL/2} r^n.$$

The series therefore converges uniformly on compact subsets of U. Each summand is analytic there, hence so is g. In particular g is analytic on a neighborhood of every point of the required closed cut disk. The choice $h = 1$, $\alpha = 1$ meets the stated hypothesis, with $h(1/2) = 1 \neq 0$.

The entire line of dominant poles. For $\operatorname{Re} s > \log 2$, the Dirichlet series satisfies

$$F(s) = \sum_{n\ge 1} 2^n e^{-s(n+a2^{-n})} = \frac{2e^{-s}}{1-2e^{-s}} + H(s),$$
$$H(s) = \sum_{n\ge 1} 2^n e^{-ns}\big(e^{-as2^{-n}} - 1\big).$$

Its abscissa is $\log 2$, by the same absolute-value test. On a compact subset of $\operatorname{Re} s > 0$, take $\operatorname{Re} s \ge \sigma > 0$ and $|s| \le M$. Now the nth summand of H has absolute value at most

$$aMe^{aM/2}e^{-n\sigma}.$$

Consequently H is analytic throughout this larger half-plane. The displayed expression continues F meromorphically across its entire convergence line. Its singularities there are exactly

$$s_k = \log 2 + 2\pi i k, \qquad k \in \mathbb{Z}.$$

At each s_k, the numerator $2e^{-s}$ and the derivative of the denominator $1 - 2e^{-s}$ both equal 1. Every pole is simple, has residue 1, and cannot be canceled by the analytic function H. This verifies the full pole condition with $\omega = 1$, without substituting across a logarithmic branch.

The asserted periodic asymptotic is impossible. Write $A(x) = |\mathcal{C}_{\le x}|$, and suppose that $A(x)/(c(x)2^x) \to 1$ for a 1-periodic c. Fix any $r \in (0,1)$. For every sufficiently large integer N, $a2^{-N} < r$, so

$$A(N+r) = \sum_{n=1}^{N} 2^n = 2^{N+1} - 2.$$

Since $c(N+r) = c(r)$, the assumed asymptotic forces

$$c(r) = 2^{1-r} \quad \text{for every } r \in (0,1).$$

At integers it similarly forces $c(0) = 1$, because $A(N) = 2^N - 2$. These conclusions require no regularity of c.

Set $\delta_N = a2^{-N-1}$ and $x_N = N + \delta_N$. Here $N < x_N < \lambda_N$, so $A(x_N) = 2^N - 2$, whereas periodicity and the already forced values on $(0,1)$ imply $c(x_N) = 2^{1-\delta_N}$. Thus

$$\frac{A(x_N)}{c(x_N)2^{x_N}} = \frac{2^N - 2}{2^{N+1}} = \frac{1}{2} - 2^{-N} \longrightarrow \frac{1}{2},$$

contradicting the required limit 1. □

Scope and verification

The failure comes from counting jumps whose phases move toward zero. The limit along each fixed fractional part exists, but those limits do not produce an asymptotic valid for all real $x \to \infty$. The example uses positive integral multiplicities, genuinely irrational sizes, a nonvanishing singular coefficient, and identical simple principal parts at every boundary pole. The source permits general locally finite size sets; it does not require the size of a word to be the sum of fixed letter weights.

The proof is self-contained. The accompanying verifier independently counts with exact arithmetic in $\mathbb{Q}(\sqrt{2})$, checks the local residue symbolically, and compares two convergent representations of the generating functions numerically. The numerical checks are supplemental; analyticity and the limiting contradiction follow from the displayed estimates and identities for all n. The complete current preprint (29 pages) and journal article (31 pages) were read. Conjecture 6 remains unchanged in the final. The separate source and literature audits record the bounded eligibility checks. Review was by the authoring agent; no independent peer review or proof-assistant certification is claimed.

Task 2: fixed lattice phases

Establish a corrected theorem for size sets contained in a fixed finite union of translates of $\omega\mathbb{N}$. Specify the coefficient asymptotics needed to make the resulting periodic approximation valid at every real counting threshold.

Task 3: smoothed counting functions

Determine which averaging kernels convert the boundary pole data into a periodic leading term. Quantify how the answer depends on the averaging width, including widths that tend to zero.

Task 4: stability under perturbing sizes

For multiplicities b^n at sizes $n + \varepsilon_n$, classify the perturbations that preserve a periodic cumulative asymptotic. Describe the exceptional intervals near jumps when such an asymptotic holds away from those intervals, and estimate their total length.

Task 5: additional analytic conditions

Identify boundary growth or uniform Tauberian hypotheses that rule out moving jumps. Pole positions and even their full principal parts alone are insufficient, as this example shows. Tasks 2–5 are proposed research directions and carry no extra credit.

References

[1] D. Bevan and J. Condé, *Introducing irrational enumeration: analytic combinatorics for objects of irrational size*, Electronic Journal of Combinatorics **33**(3) (2026), P3.15, Conjecture 6, p. 15. `https://doi.org/10.37236/14759`. Current preprint: arXiv:2412.14682v3 (22 December 2025), Conjecture 6, p. 14. `https://arxiv.org/abs/2412.14682v3`.